CSci 23: Module 1 →

Front-end Web Dev

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Course Outcome

Deploy websites using frameworks and platforms.

Learning Outcomes

1. Identify fundamental concepts on web development.
2. Develop front-end websites using frameworks.
3. Deploy websites on Platforms-as-a- Service (PaaS).

Outline

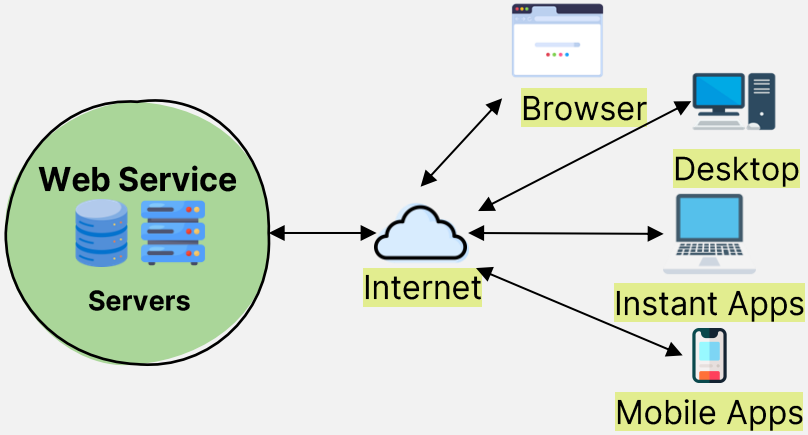
1. Concepts
2. Frameworks
3. Runtime Environments

Front-end Web Dev

Concepts

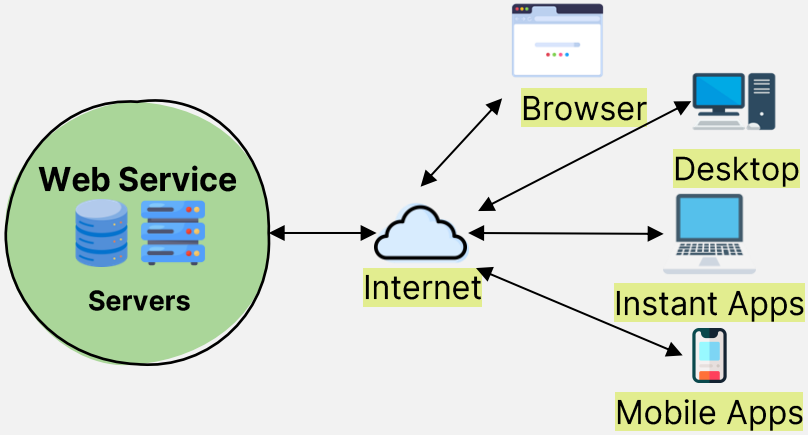
What is a Client?

It is an application or device that **requests services** or resources from another system.



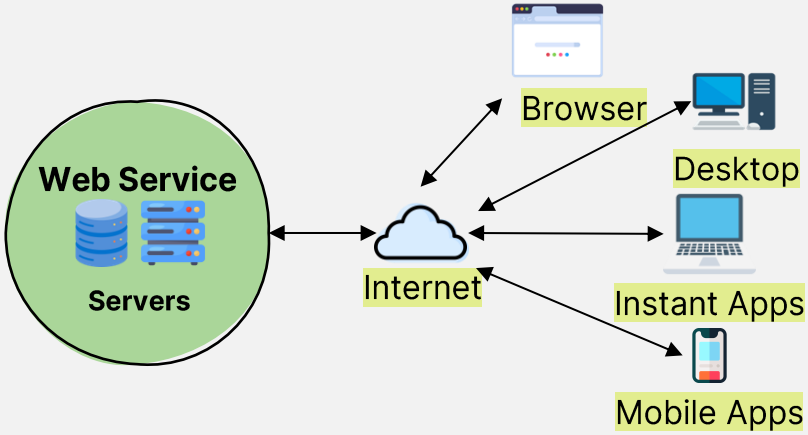
What is a Server?

It is an application or device that **provides services** or resources to clients.



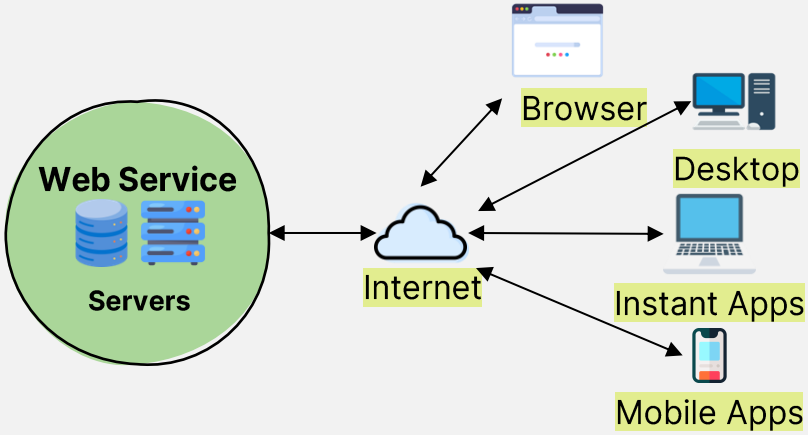
What is Front-end?

It is the **user-facing application layer** responsible for presentation and interaction, typically running on the client-side.



What is Backend-end?

It is a **server-side application layer** responsible for business logic, data processing, and database interaction.



Document Object Model

It is a programming interface representing the **dynamic structure of an HTML document in-memory**; which allows JavaScript to modify it.

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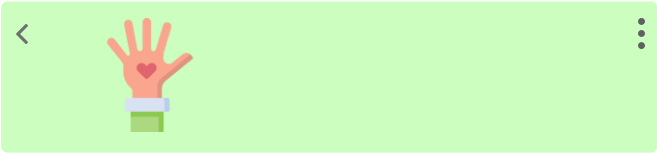
Frameworks

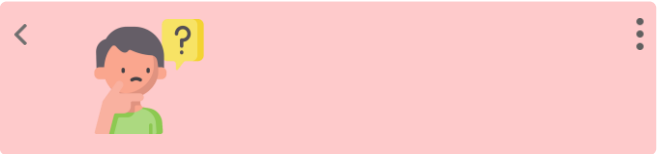
Frontend Web Frameworks

It is a set of **reusable libraries** or **packages** to help developers focus more on higher-level tasks to speed up the front-end web development process.



Pros

1. Consistency and standardization
2. Increased productivity
3. Community support
4. Scalability



Cons

1. Steep learning curve
2. Functionality constraints
3. Overhead and performance
4. Lock-in

jQuery (2006)

A lightweight JavaScript library designed to simplify HTML DOM tree manipulation, event handling, animations, and AJAX.

https://github.com/jquery/jquery

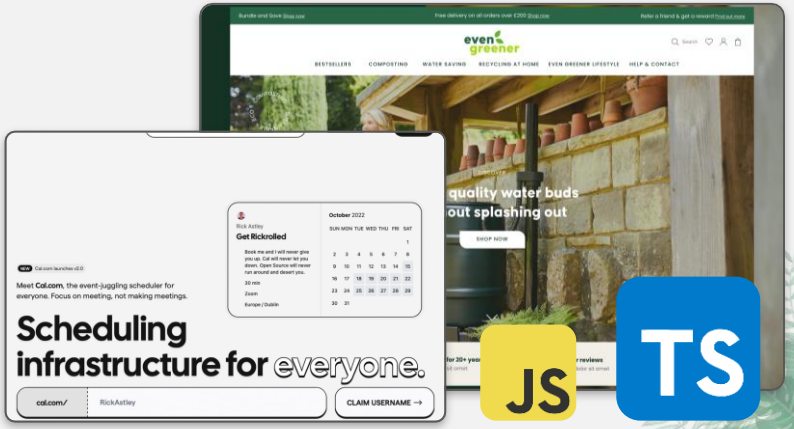
AngularJS (2010)

AngularJS is a discontinued web framework using a Model-View-Controller (MVC) architecture, focused on dynamic, single-page applications.

https://github.com/angular/angular.js

React (2013)

A declarative and lightweight web framework for building component-based and UI-heavy web applications; it uses an optimized virtual DOM.



https://github.com/facebook/react

Vue.js (2014)

It is a model–view–viewmodel (MVVM) web framework for building user interfaces and single-page applications; best for small to mid-sized web apps.

https://github.com/vuejs/core

Svelte (2019)

It is a lightweight web framework that shifts to built-in reactivity and compile-time to produce efficient vanilla JavaScript without virtual DOM overheads.

https://github.com/sveltejs/svelte

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Runtime Environments

Runtime Environment

It is a software layer that executes a program after it has been compiled or interpreted.

What is Node.js?

It is a runtime environment that allows JavaScript to run outside the browser and function like a native application, providing features such as file I/O, networking, and other system-level capabilities.

What is Node Package Manager ?

It is a package manager for Node.js used to install, manage, and share libraries and dependences.

What is Yarn?

It is an alternative to **npm** that manages dependencies at project level, ensuring consistent installations across different environments.

Platform-as-a-Service

It is a cloud computing model where a provider delivers a managed platform, including infrastructure and tools, for building and deploying applications.

Summary

- **Client** and **server** are applications and devices that interacts to request and provide data to other applications.
- **Front-end** and **back-end** are application layers residing in the client- and server-side.
- **Web Frameworks** allow faster web development.
- **Runtime Environments** allow software to be executed after being compiled or interpreted.

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Evaluation Form

forms.gle/TncdC5nEKJZYucVz6

Thank You!

References

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